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```
clc;
clear;
close all;
```

GSM TDMA SIMULATION

Four GSM Users

```
% User 1 Data
u1 = [1 0 1 1];
```

```
% User 2 Data
u2 = [0 1 0 1];
```

```
% User 3 Data
u3 = [1 1 0 0];
```

```
% User 4 Data
u4 = [0 0 1 1];
```

TDMA Frame

```
% Combine data of all four users into one TDMA frame
tdma = [u1 u2 u3 u4];
```

Display Results

```
disp('=====');
disp('      GSM TDMA SIMULATION');
disp('=====');

disp('User 1 Data:');
disp(u1);

disp('User 2 Data:');
disp(u2);

disp('User 3 Data:');
disp(u3);
```

```
disp('User 4 Data:');
disp(u4);
```

```
disp('TDMA Frame:');
disp(tdma);
```

```
=====
      GSM TDMA SIMULATION
=====
```

User 1 Data:

1 0 1 1

User 2 Data:

0 1 0 1

User 3 Data:

1 1 0 0

User 4 Data:

0 0 1 1

TDMA Frame:

Columns 1 through 13

1 0 1 1 0 1 0 1 1 1 0 0 0

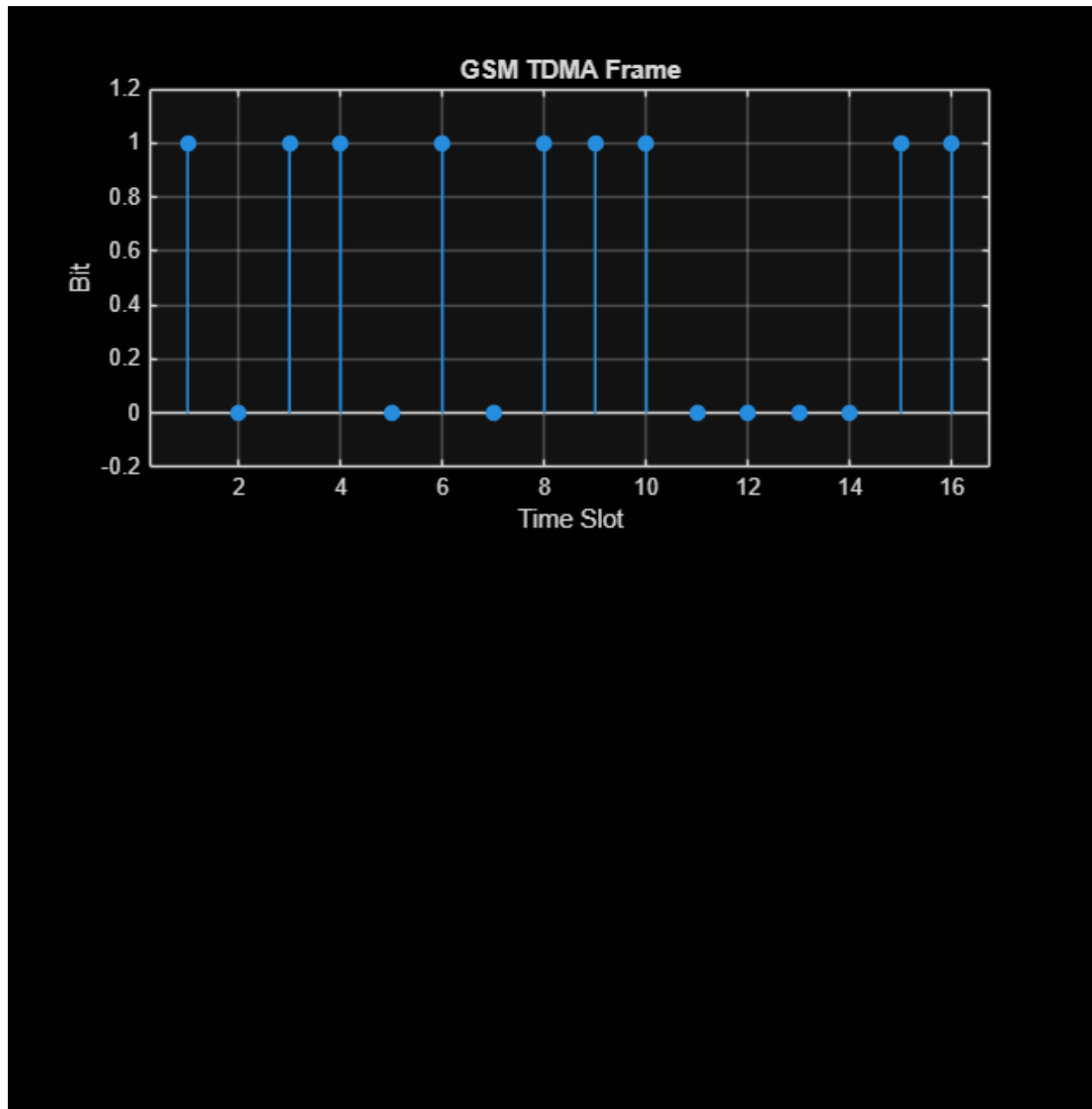
Columns 14 through 16

0 1 1

Plot TDMA Frame

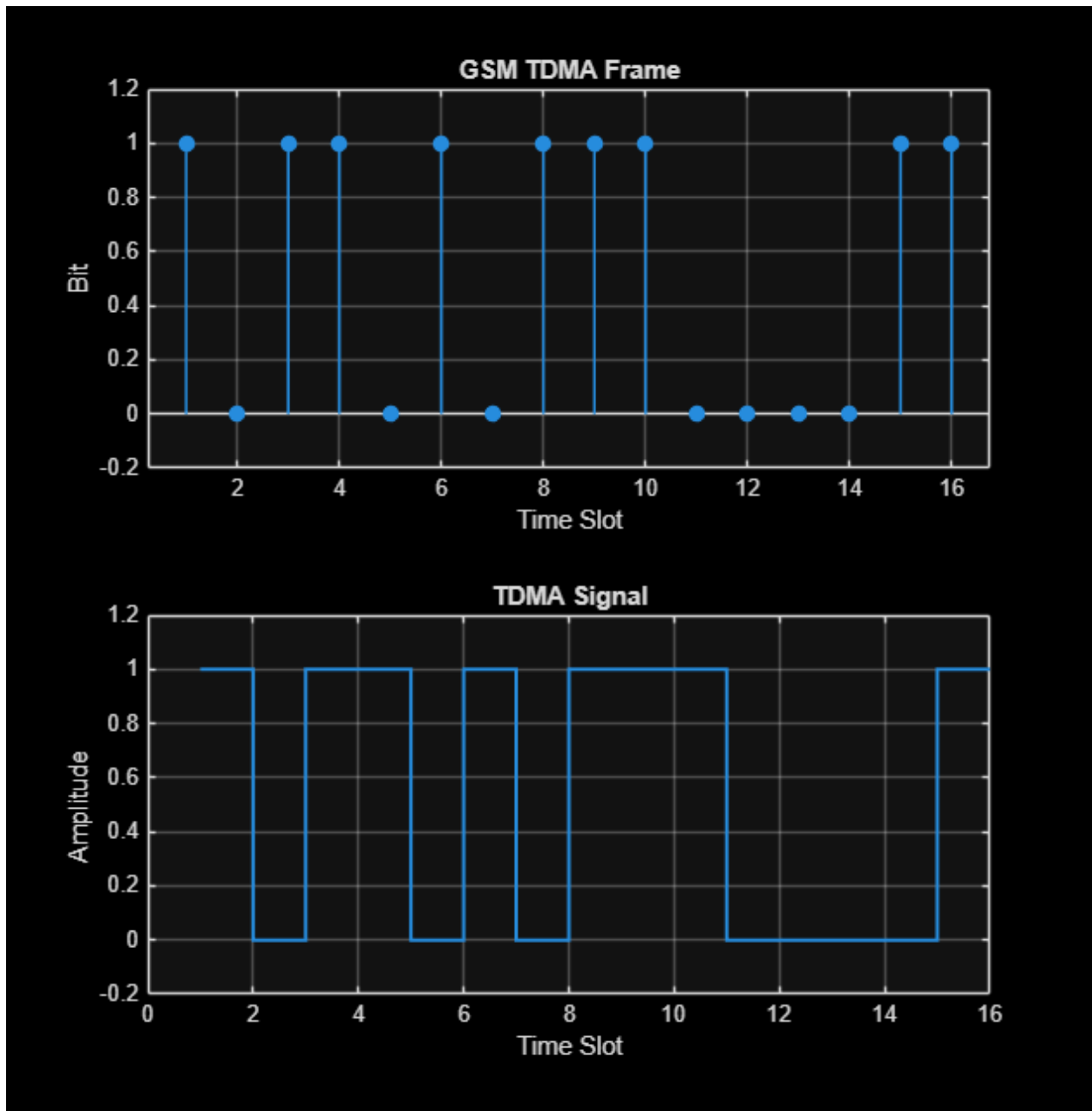
```
figure;
```

```
subplot(2,1,1);
stem(1:length(tdma), tdma, 'filled', 'LineWidth', 1.2);
title('GSM TDMA Frame');
xlabel('Time Slot');
ylabel('Bit');
ylim([-0.2 1.2]);
grid on;
```



Plot TDMA Signal

```
subplot(2,1,2);  
stairs(1:length(tdma), tdma, 'Linewidth', 1.5);  
title('TDMA Signal');  
xlabel('Time Slot');  
ylabel('Amplitude');  
ylim([-0.2 1.2]);  
grid on;
```



Display User Time Slots

```
fprintf('\n=====\\n');
fprintf('          TIME SLOT ALLOCATION\\n');
fprintf('=====\\n');

fprintf('User 1 -> Time Slots 1 to 4\\n');
fprintf('User 2 -> Time Slots 5 to 8\\n');
fprintf('User 3 -> Time Slots 9 to 12\\n');
fprintf('User 4 -> Time Slots 13 to 16\\n');

fprintf('\\nTotal Number of Time Slots = %d\\n', length(tdma));
```

```
fprintf('Total Number of Users = 4\n');  
  
disp('=====');  
disp('Simulation completed successfully.');
```

```
=====
                        TIME SLOT ALLOCATION
=====
User 1 -> Time Slots 1 to 4
User 2 -> Time Slots 5 to 8
User 3 -> Time Slots 9 to 12
User 4 -> Time Slots 13 to 16  
  
Total Number of Time Slots = 16
Total Number of Users = 4
=====
Simulation completed successfully.
=====
```

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